Jiagen Geng

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RESEARCH INTERESTS

Fate and treatment of emerging contaminants in natural and engineered water systems, with a current focus on tire wear particle (TWP)-derived compounds such as 6PPD and its transformation product 6PPD-quinone, a growing regulatory concern for salmon-bearing waters. Current work pairs electrocoagulation and activated carbon adsorption with electrochemical characterization and computational modeling to understand how these contaminants interact with sorbents. Earlier work addressed antibiotics in agricultural systems and their degradation by advanced oxidation.

EDUCATION

Michigan State University — **East Lansing, US**
*Ph.D. Student in Environmental Engineering (*Advisor: Dr. Shiqiang Zou*)* — Aug 2025 – Present
- Honors: *Clifford Humphrys Fellowship* — 2026–2027
  Certificate of Merit, ACS Division of Environmental Chemistry — 2026
- Relevant Coursework: Quantum and Atomistic Simulations for Materials

Auburn University — **Auburn, US**
*Ph.D. Student in Civil Engineering (*Advisor: Dr. Shiqiang Zou*)* — Aug 2023 – Aug 2025
- Honors: *Charles E. Gavin Fellowship* — 2023–2025
- Relevant Coursework: Electrified Water Treatment, Big Data Analytics and Machine Learning in Process Industry, Chemical Engineering Thermodynamics

China Agricultural University — **Beijing, China**
Master of Environmental Science and Engineering — Sep 2021 – Jun 2023
- Honors: First-class, Second-class, and College Scholarships
- Relevant Coursework: Environmental Fate and Control of Emerging Contaminants, Environmental Pollution Chemistry, Environmental Microbiology

Hebei University of Technology — **Tianjin, China**
Bachelor of Environmental Engineering — Sep 2017 – Jun 2021
- Honors: First-class, Second-class, and College Scholarships
- Relevant Coursework: Water Pollution Control Engineering, Air Pollution Control Engineering, Process and Systems Engineering, Energy Management

EXPERIENCE

Michigan State University — **East Lansing, US**
Research Assistant — Aug 2025 – Present
- Investigated electrochemical and adsorptive treatment of tire wear particle (TWP)-derived contaminants in stormwater, including electrocoagulation and granular activated carbon adsorption
- Performed density functional theory calculations (ORCA, CP2K) on adsorbent models to interpret interaction behavior
- Developed HPLC and LC-MS/MS methods for trace-level quantification of 6PPD and 6PPD-quinone; built Python workflows to process thousands of electrochemical datasets

Teaching Assistant

- ENE 480, Environmental Measurements Laboratory (Fall 2025): supervised laboratory sessions and graded lab reports

Auburn University **Auburn, US**

Research Assistant Aug 2023 – Aug 2025

- Designed and executed electrochemical water treatment experiments, including cell assembly, protocol development, and system operation
- Operated Bio-Logic potentiostats for cyclic voltammetry, electrochemical impedance spectroscopy, and galvanostatic charge/discharge measurements

Teaching Assistant

- CIVL 3230, Introduction to Environmental Engineering (Spring 2024, Fall 2024, Spring 2025): delivered laboratory lectures covering experimental theory and procedures, supervised weekly lab sections, held office hours, and graded assignments

U.S. EPA-Funded Research Projects

Project Member Oct 2026 – Present

- Project: *Carbon-Loaded Electrochemical Artificial Retention (CLEAR) System for Nutrient Reduction and Recovery in Agricultural and Urban Runoff*
- Contributed to the Quality Assurance Project Plan (QAPP) and coordinated activities across a multi-institution student team

Project Member Aug 2023 – May 2026

- Project: *Non-woven hybrid geotextiles for urban stormwater nutrient management*
- Conducted field sampling and bench-scale tests
- Contributed to the QAPP and prepared monthly progress reports

China Agricultural University **Beijing, China**

Research Assistant Sep 2021 – Jun 2023

- Investigated biochar-based adsorption for removal of antibiotic contaminants from agricultural wastewater, including biochar synthesis by pyrolysis
- Systematically reviewed the occurrence and risks of antibiotics across environmental media, including rivers, agricultural soils, and crops

PUBLICATIONS & PRESENTATIONS

- **Journal Articles:**

[1] **Jiagen Geng**, Duo Wang, D. Ricardo Martínez-Vargas, Ao Xie, Aline Christine Bernegossi, Shiqiang Zou, 2026. *Structure-Dependent Selective Adsorption and Removal of 6PPD and 6PPD-quinone in Chloride-Enabled Aluminum Electrocoagulation*. Journal of Hazardous Materials 515, 143151.

[2] Huijuan Li, Zhu Wang, **Jiagen Geng**, Ruiping Song, Xiaoyin Liu, Chaochen Fu, Si Li, 2025. *Current advances in UV-based advanced oxidation processes for the abatement of fluoroquinolone antibiotics in wastewater.* Chinese Chemical Letters 36(4), 110138.

[3] **Jiagen Geng**, Xiaoying Liu, Jie Wang, Si Li, 2022. *Accumulation and risk assessment of antibiotics in edible plants grown in contaminated farmlands: A review*. Science of the Total Environment 853, 158616.

[4] Si Li, Yi Liu, Yang Wu, Jingrun Hu, Yiqing Zhang, Qian Sun, Weiling Sun, **Jiagen Geng**, Xiaoying Liu, Dantong Jia, Xiuqi You, Dianqing Qi, Moran Tang, Yitao Lyu, Fanguang Kong, Leilei Cai, Yufan Ai, Yichu Wang, Jinren Ni, 2022. *Antibiotics in global rivers*. National Science Open 1(2), 20220029.

[5] Xiaoying Liu, Ying Li, **Jiagen Geng**, Jie Wang, Qing Chen, Si Li, 2023. *Effects of different fertilization regimes on antibiotics and antibiotic resistance genes in greenhouse vegetable soil*. China Environmental Science 43(2), 772–780. (in Chinese)

- **Conference Paper**:

[6] **Jiagen Geng**, Fanguang Kong, Si Li, 2022. *Activation of peroxymonosulfate by composite material of cobalt ferrite and biochar for norfloxacin degradation*. Proceedings of the 1st International Symposium on Urban and Rural Non-point Source Pollution Control, Nanjing, China. (in Chinese)

- **Presentations**:

[7] (Oral) **Jiagen Geng**, D. Ricardo Martínez-Vargas, Shiqiang Zou. *Deciphering electrochemical carbon fingerprints: A case study with 6PPD and 6PPD-quinone*, ACS Fall Meeting, Chicago, IL, 2026.

[8] (Poster) **Jiagen Geng**, D. Ricardo Martínez-Vargas, Shiqiang Zou. *Probing 6PPD and 6PPD-quinone adsorption on activated carbon via electrochemical fingerprinting*, Borchardt Conference, Ann Arbor, MI, 2026.

[9] (Oral) **Jiagen Geng**, Duo Wang, D. Ricardo Martínez-Vargas, Ao Xie, Aline Christine Bernegossi, Shiqiang Zou. *Removing 6PPD and 6PPD-quinone from stormwater runoff with an electrocoagulation system*, ACS Spring Meeting, San Diego, CA, 2025.

[10] (Poster) **Jiagen Geng**, D. Ricardo Martínez-Vargas, Ao Xie, Aline Christine Bernegossi, Shiqiang Zou. *Electrocoagulation-aided removal of tire wear particle-derived pollutants in stormwater runoff*, AEESP Conference, Durham, NC, 2025.

[11] (Poster) Aline Christine Bernegossi, **Jiagen Geng**, Shiqiang Zou. *Evaluating the resistance of Rudbeckia hirta and Rumex crispus to 6PPD-quinone and potential use for runoff phytoremediation*, AEESP Conference, Durham, NC, 2025.

[12] (Poster) **Jiagen Geng**, Duo Wang, D. Ricardo Martínez-Vargas, Ao Xie, Aline Christine Bernegossi, Shiqiang Zou. *Electrocoagulation Treatment for the Removal of 6PPD and 6PPD-Quinone from Stormwater Runoff*, AL/MS Water Conference, Mobile, AL, 2025.

[13] (Poster) **Jiagen Geng**, Megan Armstrong, Michael Perez, Shiqiang Zou. *Electrocoagulation of the Tire Wear Particle-derived 6PPD & 6PPD-quinone in Stormwater Runoff*, AL/MS Water Conference, Mobile, AL, 2024.

PROFESSIONAL SERVICE & MEMBERSHIPS

Journal Reviewer: *Desalination and Water Treatment* (3 manuscripts)
Member, American Chemical Society (ACS)

SKILLS

Environmental analysis: HPLC, LC-MS/MS (Agilent 1100/1260/1290, Waters triple quadrupole), UV-Vis, ELISA, IC, TOC
Electrochemistry: Bio-Logic SP-300 and VMP-300 potentiostats (EC-Lab): CV, LSV, EIS, CA, CP, GCD
Materials characterization: FTIR, EPR
Programming: Python, MATLAB, R
Computational chemistry: DFT (ORCA, CP2K), molecular dynamics (Materials Studio)
Process modeling & design: SUMO, AutoCAD
Field & quality assurance: field sampling, QA/QC, QAPP development
Languages: Mandarin (native), English (fluent), German (basic)